Joshua Walker

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EDUCATION

Imperial College London 2022 — 2026

MEng Computing - graduated with **First Class honours**. Formative modules included:

- 60015 **Network and Web Security** (72.80%) – learnt **pentesting** methods and exploitation of **CVEs**;
- 60005 **Graphics** (69.10%) – studied graphics pipelines, investigated shading and illumination using **GLSL**;
- 70001 **Advanced Graphics** (80.46%) – investigated **photorealism** using subsurface-scattering models to accurately render skin, **relit scenes** using HDR environment light probes, and learnt practical methods used in cinematic VFX from an industry veteran.

EMPLOYMENT HISTORY

Elliott Management — Junior Software Engineer Apr 2025 — Sep 2025

Interned at London branch of the New York-based activist hedge fund. Key projects included:

- Providing **traders** and **portfolio managers** with automated reports to aid market assessments;
- Automating systems to **calculate** and **standardise derivative prices** across multiple critical datasets;
- Restructuring existing solutions for monitoring support systems into a **robust** and **consolidated** framework.

PROJECTS

Ginseng (Final Year Project, 65.00%) – ScalaNative, OpenGL Jan 2026 — Jun 2026

Proposed and developed a project utilising principled **staged metaprogramming techniques** to efficiently convert from an ergonomic functional programming Domain Specific Language for rendering graphics into standardised OpenGL (via a **bespoke** ScalaNative bindings library developed alongside Ginseng).

Dill (Group Project, 82.40%) – Rust, ScalaJS, Tauri, Laminar, CSS Jan 2025 — Mar 2025

Built an **interactive cross-platform debugger** for the parser combinator library Parsley (utilised in Imperial's compilers coursework), working throughout semester to deliver an app meeting both the brief of our supervisor and the needs of our fellow undergraduates.

Designing for Real People (Group Project) – Flutter, Firebase, Dart, CSS Jun 2024

Developed a recipe-sharing social media for students, using Human-Centered Design techniques taught by the **Royal College of Art** to ensure commercial viability and meet the demand of our target demographic.

Pintos (Group Project) – C Oct 2023 — Dec 2023

Extended a simple operating system with **synchronisation primitives**, **system calls** and **virtual memory** to explore the implementation of foundational OS concepts within a pre-existing codebase.

Origame (A Level Project, A-A*) – Unity, C# Jan 2022 — May 2022

Developed a 2D puzzle game in the Unity engine, documenting the design of a **platform-folding** mechanism.

TECHNICAL SKILLS

Languages: Scala, Rust, C, C#, Java, Haskell, OpenGL/GLSL, Python, JS, HTML, CSS, Assembly, Git, LaTeX

EXTRACURRICULAR

Backpacked around the world to **5 continents** in my graduate summer — hiked the **Great Wall** in Beijing, Japan's iconic **Mt. Fuji**, and the 7-day **Salkantay-Inca Trail** to Machu Picchu.

Self-taught **guitarist**. Member of **Muay Thai Society**. Aspiring **sketch artist** – studied the Practical Art module provided at Imperial College by a former lecturer and PhD student at Goldsmiths.